nature communications

Article | Article in Press

https://doi.org/10.1038/s41467-026-77794-5

Van der Waals epitaxy suppresses thermal rupture in ultrathin metals

Received: 11 May 2026

Accepted: 1 September 2026

Published online: 16 September 2026

Check for updates

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Metal films are indispensable to modern electronics, yet their continued downscaling is fundamentally limited by thermodynamic instability. In particular, ultrathin metals can exhibit liquid-like Plateau–Rayleigh instability and rupture at temperatures far below their bulk melting point, creating a long-standing challenge for both integrated circuits and emerging 2D-material devices. Here we show that the liquid-like instabilities of thin metals can be suppressed at the microstructural origin by using graphene as a van der Waals (vdW) template. The vdW template induces a highly uniform texture and reorganizes grain boundaries into a low-energy network, thereby shifting capillary driving forces from rupture-promoting to self-healing. Remarkably, while conventional 20-nm-thick gold films break up into islands below 300 °C, templated films not only remain continuous above 600 °C but also evolve into atomically flat structures. We show that this exceptional stability greatly broadens the processing window for nanoscale interconnects and electrodes, and enables high-quality integration of metals with 2D materials.

Dewetting is typically associated with liquids: thin films on non-wetting surfaces spontaneously retract and break up into droplets via Plateau–Rayleigh instability[1]. Yet solids can show strikingly similar behavior[2–4]. At nanometer scales or elevated temperatures, both crystalline and amorphous solid films develop liquid-like instabilities that drive roughening, rupture, and eventual loss of continuity[2–5]. Metal films are especially susceptible[6]. In Au, for example, the as-deposited microstructure is far from equilibrium, rendering the film highly prone to rupture[7]. The thinner the film, the lower the rupture temperature, which can fall to as little as one-tenth of the bulk melting point (T_m)[7].

Although this instability has long plagued makers of conventional integrated circuits and microsystems[7,8], it has become even more problematic in 2D-material electronics. In such systems, ultrathin metals (< 50 nm) serve as contacts, gates, and interconnects, while thermal annealing is essential for forming low-resistance metal–2D junctions[9–24]. As a result, fabrication protocols are typically confined to temperatures below 300 °C[9–24], well below the ~400 °C thermal budget commonly required for standard semiconductor integration[14,15,25].

In this work, we show that vdW epitaxy can overcome this limitation by modulating the microstructure of ultrathin metals to achieve extraordinary thermal stability. The vdW template promotes a highly textured Au microstructure with a low-energy grain-boundary network, suppressing rupture and favoring surface levelling during annealing. This stabilization substantially expands the thermal processing window of ultrathin metals and enables robust metallic components for high-temperature device integration.

Results

From rupturing to capillary healing

Annealing conditions for representative metal–2D-material devices are mostly confined to temperatures below 300 °C[9–24] (Supplementary

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https://doi.org/10.1038/s41467-026-77794-5

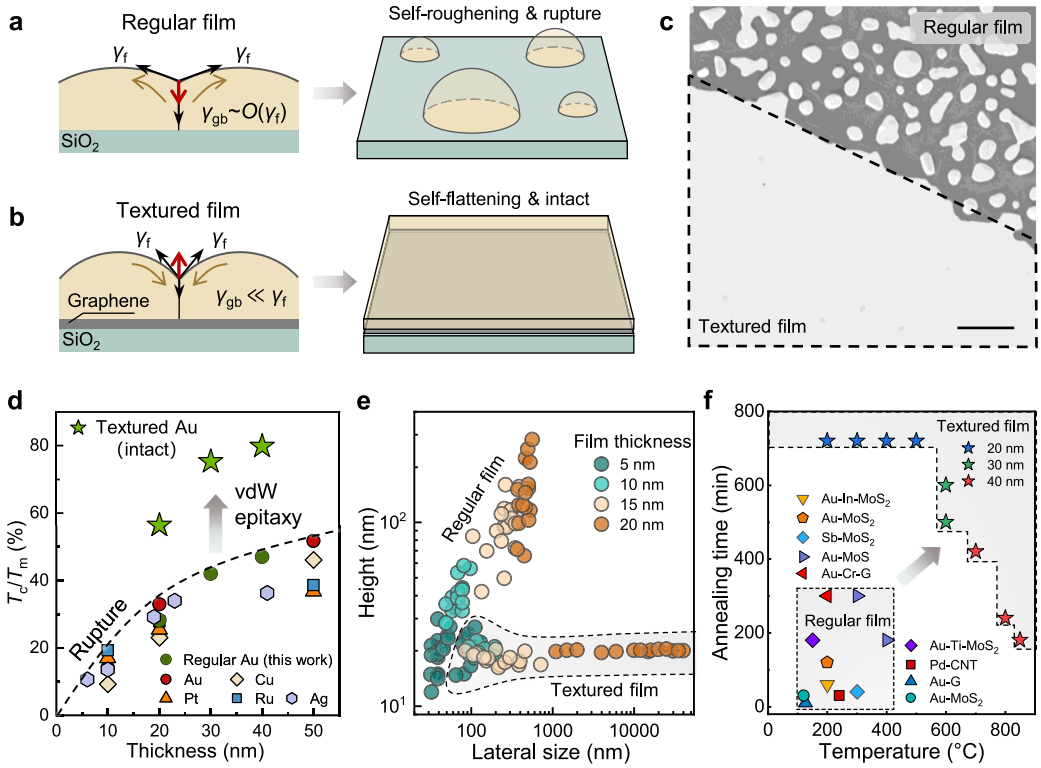


Fig. 1 | Evolution of solid films at elevated temperatures. a Schematic of a conventional polycrystalline metal film (regular film), in which surface diffusion drives self-roughening and ruptre. γ_f and γ_{gb} denote the film surface energy and grain-boundary energy, respectively; arrows indicate the direction of capillary-driven surface flow. **b** Schematic of a graphene-templated film (textured film), where surface flow is directed toward self-flattening rather than breakup. **c** Scanning electron microscope (SEM) image of a 20-nm-thick Au film annealed at 600 °C for 2 hours; the dashed box highlights the textured region. Scale bar: 2 μm. **d** Critical rupture temperatures T_c for conventional metal films (lower markers; data reproduced from ref. 7.) and temperatures at which textured Au films remain intact (upper markers). All temperatures are normalized by the melting point T_m. **e** Height and lateral dimensions of Au films after annealing at 600 °C for 2 hours. Note that textured films (in shaded area) only break up at small thicknesses and begin to flatten when their thickness exceeds 20 nm, such that the lateral size is controlled by the size of the underlying graphene sheet (>1 μm). **f** The lower-left shaded area indicates the annealing window reported for representative 2D material devices in the literature, including Au-In with MoS_2[9], Au with MoS_2[10], Sb with MoS_2[14], atomic layer bonding (Au-MoS_2)[15], Au with 2D materials with Cr or Ti adhesion layer[18], Pd with carbon nanotubes[19], Au with graphene[20], and transferred Au with MoS_2[24]. Star markers represent specific conditions where textured Au films are found to remain intact, and the upper-right shaded area highlights the annealing window sustained by textured Au films in this work.

Fig. 1), highlighting the need for ultrathin metal films with improved thermal stability. Motivated by this limitation, we compared conventional Au films with films grown on a van der Waals template. Au films with thicknesses of 5–40 nm were fabricated by standard electron-beam evaporation (see Methods). Films deposited directly onto SiO_2 substrates are referred to as "regular" films, while those templated onto SiO_2 coated with mechanically exfoliated graphene are termed "textured" films. Atomic force microscopy (AFM) reveals that the textured films exhibit slightly higher roughness than the regular films (Supplementary Fig. 2), as illustrated in Fig. 1a, b. This behavior aligns with prior reports on Au on 2D crystal surfaces[26], and may reflect subtle changes in effective surface energy introduced by the atomically thin coating[27].

As deposited, both types of film are far from equilibrium and therefore evolve upon heating[7]. Yet their evolution is fundamentally different. Scanning electron microscopy (SEM) (Fig. 1c and Supplementary Fig. 3) together with AFM (Supplementary Fig. 3) shows that, after annealing at 600 °C for 2 h, 20-nm-thick regular films dewet and rupture into isolated islands, whereas textured films remain continuous and, strikingly, become flatter than in their as-deposited state.

The rupture of regular films follows a well-established failure mode in metallic thin films[7]. For ultrathin metals such as Au, Ag, and Cu with thicknesses of 5–50 nm, the critical failure temperature typically lies between ~10% and ~50% of T_m (Fig. 1d), consistent with our measurements[28,29]. Mullins showed that an ideal planar metallic film is intrinsically stable to perturbations[30]. However, real films are polycrystalline, and, as Srolovitz and Safran later pointed out, they almost inevitably form grooves where grain boundaries meet the free surface (Fig. 1a)[31]. In this configuration, disjoining pressure plays little role, whereas the grain-boundary energy γ_{gb} can be comparable to surface energy γ_f, generating a capillary driving force that deepens the groove. Surface diffusion then transports material away from the groove, nucleating holes and eventually leading to rupture at relatively low temperatures[32].

By contrast, the flattening of textured films is quite counterintuitive. These films are initially slightly rougher than regular films (Supplementary Fig. 2), and the atomically smooth graphene surface is expected to reduce interfacial friction[33]; both effects should enhance surface diffusion and favor rupture[34]. Consistent with this expectation, previous studies found that ultrathin Au films (<5 nm) develop more mature island patterns on graphene than on SiO_2[35]. Our measurements likewise show that textured films still break up below ~10 nm (Supplementary Fig. 3), although the resulting islands are shallower and more puddle-like than those formed by regular films (Fig. 1e). Remarkably, once the textured films exceed ~20 nm, they no longer rupture at 600 °C, but instead evolve toward an increasingly levelled morphology (Fig. 1e). This thickness-dependent transition originates from the progressive coalescence of Au islands during growth on graphene (Supplementary Fig. 2). With increasing thickness, the as-

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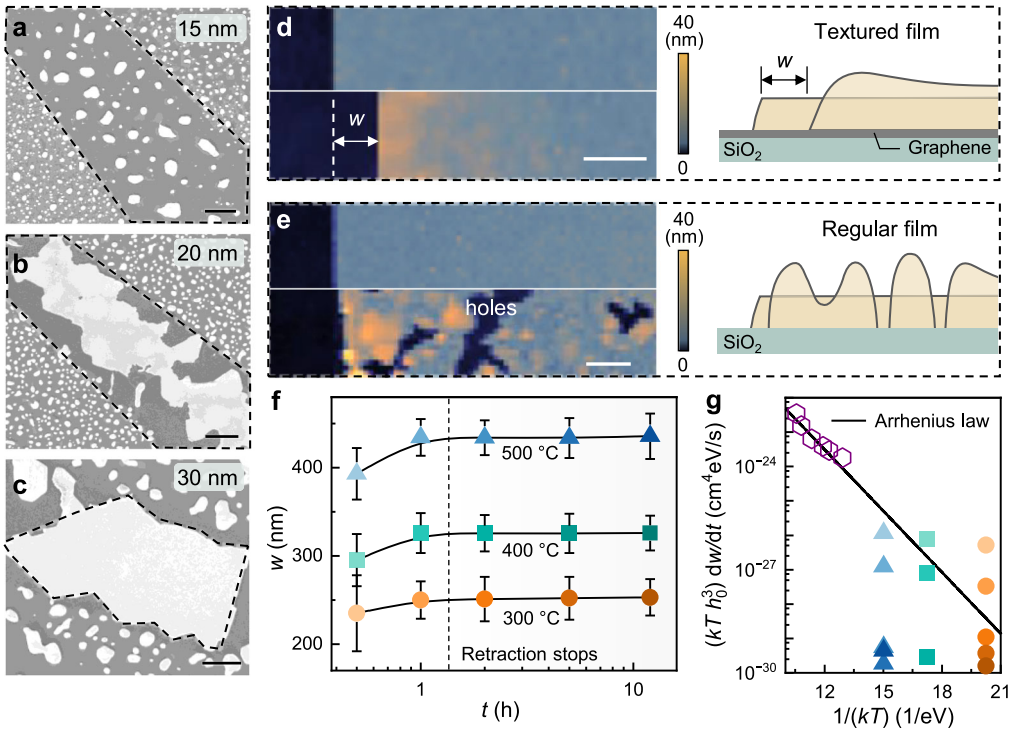


Fig. 2 | Evolution of stepped Au films. a–c SEM images of Au films with different thicknesses after annealing at 800 °C for 2 h; dashed boxes denote textured regions. **d, e** Schematic illustrations and atomic force microscope images of a stepped textured (**d**) and regular (**e**) film before and after annealing at 300 °C for 30 min, where w denotes the edge retraction length. **f** Edge retraction length w of the stepped textured film as a function of time, measured at different annealing temperatures. Error bars represent the standard deviation of edge-retraction lengths measured at five different positions on the same sample. **g** Relation between the retraction speed and annealing temperature, where k is the Boltzmann constant, T is the absolute temperature, and h_0 is the initial film thickness. Colors in (**f**) and (**g**) correspond to the annealing temperatures indicated in (**f**).The solid line and open markers represent data for regular Au films adapted from ref. 40. Scale bars: 2 µm (**a**), 5 µm (**b**), 3 µm (**c**), 500 nm (**d**, **e**).

deposited films gradually evolve from discontinuous island structures into a continuous film. The formation of such a continuous film is a prerequisite for the exceptional thermal stability subsequently exhibited by the textured Au films.

The anomalous levelling behavior allows 40-nm-thick textured Au films to remain fully intact at 850 °C for 3 h, approaching 80% of T_m (Fig. 1d and Supplementary Fig. 4e). This thermal stability is extraordinary: even 50-nm-thick single-crystal Si films begin to rupture at 800 °C after only 2 min[36], despite the melting temperature of Si being approximately 1.33 times that of Au. The stabilizing effect of vdW texture extends to thinner films. 20-nm-thick textured films are intact at 600 °C and, even at 800 °C, display only partial edge retraction while maintaining continuity across the interior (Fig. 2b). Such excellent thermal stability is not limited to monolayer graphene templates: similar behavior is observed when using other 2D material templates, including exfoliated multilayer graphene, MoS_2, and large-area chemical vapor deposition (CVD) grown graphene (Supplementary Figs. 4-5).

Because thin-metal contacts, gates, and interconnects are vulnerable to thermal failure[37], most 2D-device fabrication protocols are confined to a narrow thermal budget—typically below 300 °C and for annealing times no longer than ~300 min (Fig. 1f)[9–24], which is incompatible with standard semiconductor integration[14,15,25]. A notable exception is a recently reported atomic layer bonding approach, in which removal of the top chalcogen layer from MoS_2 permits annealing up to ~400 °C[15]. Our textured metallic films far exceed this limit, extending the accessible process window to temperatures approaching 800 °C and annealing times of ~800 min (Fig. 1f), while remaining continuous and even showing improved microstructure and electrical conductivity (to be shown shortly).

Suppressed edge retraction

Notably, textured films also remain thermally stable at their edges (Fig. 1c). This behavior contrasts with the sustained edge recession commonly observed in solid films, whose retraction kinetics have been widely used to determine solid-state diffusivities[38]. To clarify this behavior, we fabricated 20-nm-thick stepped Au films (Fig. 2d, e) and monitored edge evolution during annealing at 300 °C[39]. We find that textured films exhibit only slight edge retraction at early times (Fig. 2d and Supplementary Fig. 6), whereas regular films show both retraction and rupture near the edge (Fig. 2a–c). Remarkably, edge retraction in textured films ceases entirely after about 1 hour, as quantified in Fig. 2f.

The retraction length w of textured films evolves with time t in a way that differs qualitatively from previous measurements on thicker regular films (with unperturbed thickness $h_0 \geq 40$ nm). In the early stage ($t < 1$ h), the temperature dependence departs from the Arrhenius relation $dw/dt \propto \exp(-E_a/kT)/kT$ reported by Jiran and Thompson (Fig. 2g), where k is Boltzmann's constant, T is absolute temperature, and E_a is the activation energy[40]. Retraction also arrests too rapidly for the steady-state scaling $w \propto t^{2/5}$ previously observed in regular films to emerge[38,41].

Importantly, this arrest cannot be attributed to the graphene interface alone. A regular Au film first deposited on SiO_2/Si and subsequently transferred onto graphene still ruptures upon annealing (Supplementary Fig. 7). Conversely, a graphene-templated textured film remains continuous after graphene removal, careful poly(methyl methacrylate) (PMMA)-assisted transfer onto bare SiO_2/Si, and annealing at 600 °C for 2 h (Supplementary Fig. 8). The comparable transfer procedures, together with the preserved morphology after transfer (Supplementary Fig. 9), show that transfer-induced damage is not responsible for the contrasting thermal behaviors. We therefore

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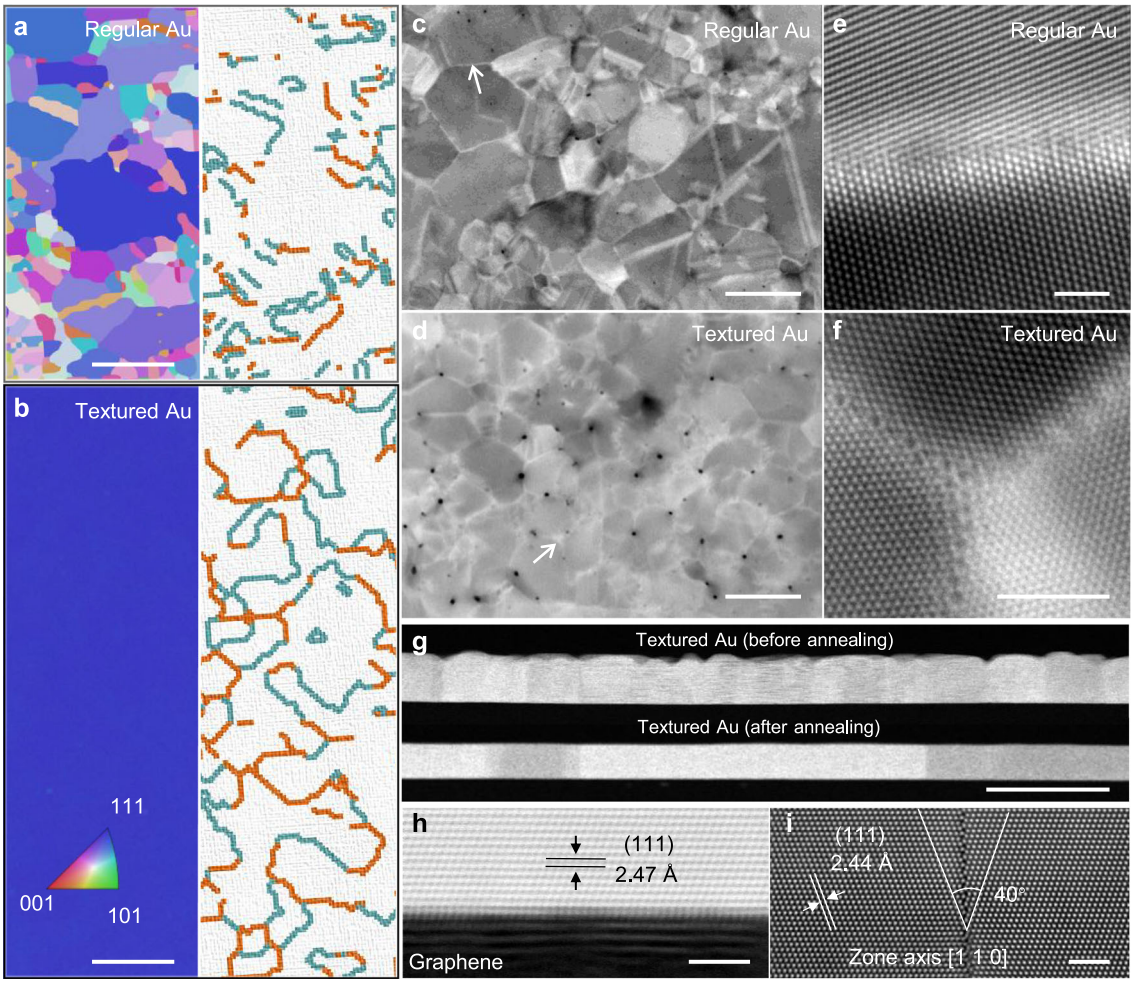


Fig. 3 | Microstructural characterization of Au films. a, b Electron backscatter diffraction (EBSD) images showing grain orientation and grain-boundary features of regular (**a**) and textured (**b**) Au films. Grain orientations are color-coded according to the inverse pole figure key; Σ3 boundaries are highlighted in green and other coincident site lattice boundaries in orange. **c, d** Plane-view high-angle annular dark-field scanning transmission microscope (HAADF-STEM) images of regular (**c**) and textured (**d**) Au films; arrows mark the grain boundaries enlarged in (**e**) and (**f**), respectively. **e, f** High-magnification plan-view images of representative grain boundaries in regular (**e**) and textured (**f**) films. **g** Cross-sectional HAADF-STEM images of a 30 nm thick textured Au film before and after annealing at 600 °C. **h** Cross-sectional images of the textured Au/graphene interface after annealing. **i** Atomic-resolution cross-sectional image of a Σ3(112) twin boundary in the textured Au film after annealing. Scale bars: 100 nm (**a**–**c**, **g**), 50 nm (**d**), 1 nm (**e**), 2 nm (**f**, **h**, **i**).

attribute the exceptional thermal behaviour of textured films primarily to their distinctive as-deposited microstructure, which we examine below.

A low-energy grain boundary network

Transmission electron backscatter diffraction (EBSD) reveals a striking contrast in crystal texture between the two classes of Au film (Fig. 3a, b). Regular films are composed of grains with largely random orientations, whereas textured films exhibit a strong out-of-plane (111) texture established during deposition by the graphene template (Supplementary Fig. 10). In addition, regular films span a broad range from 10 to 190 nm, with a strong bias towards 10–30 nm grains (Supplementary Fig. 11a). By contrast, textured films on monolayer and multilayer graphene both show narrower distributions, centred at approximately 20–30 nm and 35–60 nm, respectively (Supplementary Fig. 11a). Molecular dynamics (MD) simulations further reveal that Au atoms have much longer surface diffusion lengths on graphene than on SiO_2, promoting (111) growth and reducing grain density (Supplementary Fig. 12). In this way, the van der Waals interaction with graphene directs the formation of a uniformly textured epitaxial architecture that is intrinsically robust against thermal perturbation.

The structural uniformity of textured films produces a distinctive network of mechanically robust grain boundaries (Fig. 3a, b). Around 40% of these boundaries are low-angle and therefore low-energy, while the remaining high-angle boundaries are composed almost entirely of coincidence-site lattice (CSL) boundaries with superior atomic registry (Supplementary Fig. 11b)[42–44]. As a result, nearly all grain boundaries in textured films are low-energy interfaces, which enhances their mechanical stability. Regular films are markedly different: only ~20% of their grain boundaries are low-angle (Supplementary Fig. 11b), and although CSL boundaries are present, they occur only as short, disconnected segments rather than a continuous network (Fig. 3a). This heterogeneous microstructure leaves roughly half of all boundaries as high-energy, large-angle interfaces (Supplementary Fig. 11b), providing rapid diffusion pathways under thermal perturbation. This stark contrast leads to a net capillary force at the grain-boundary grooves of opposite directions, which drives rupture in regular films (Fig. 1a) but promotes levelling and healing in textured films (Fig. 1b).

We further examined the differences in grain size and orientation between regular and textured films by plane-view, aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) (Fig. 3c–f). In agreement with the

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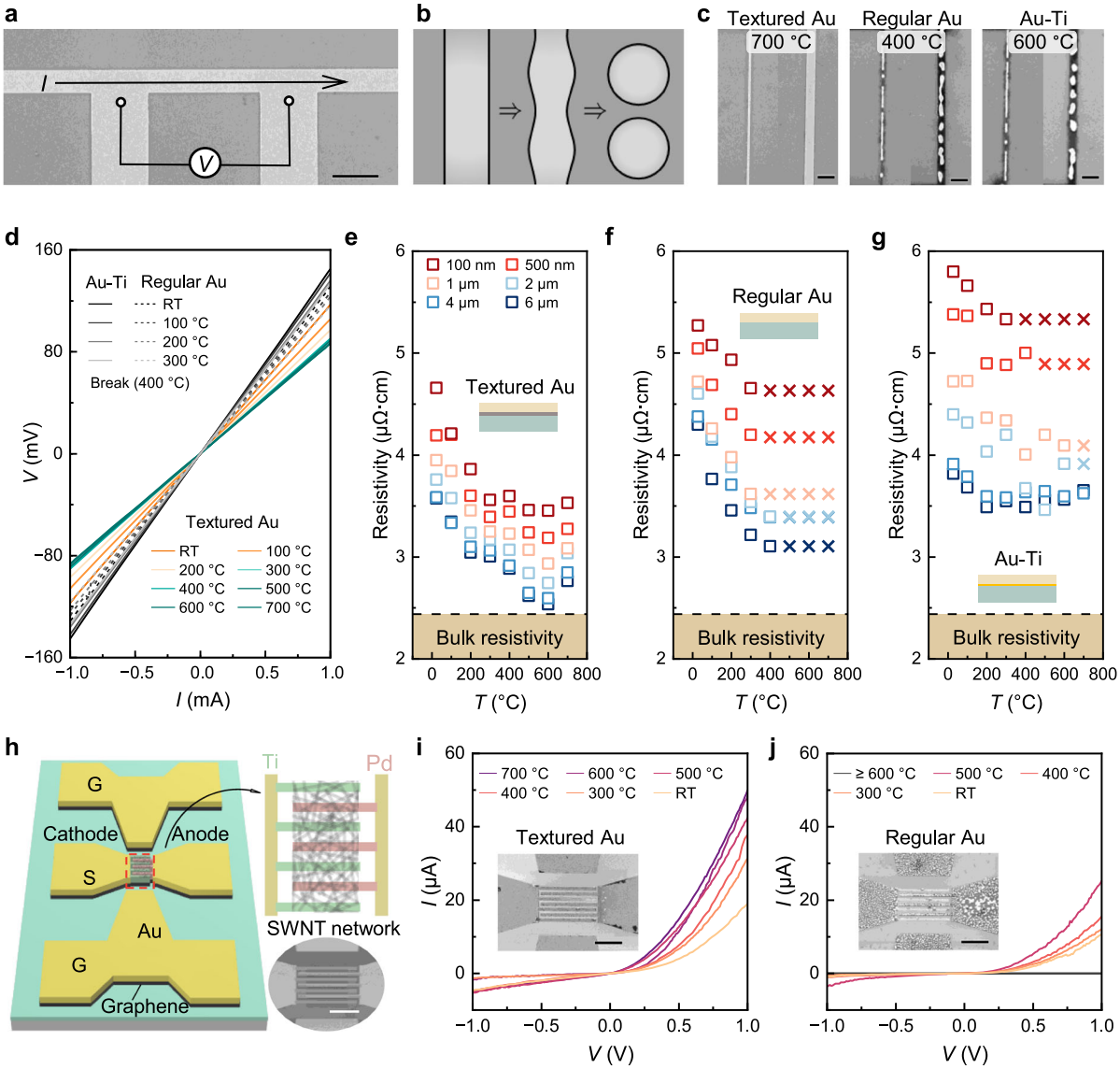


Fig. 4 | Electrical performance of textured Au ribbons. a Four-terminal measurement configuration overlaid on an scanning electron microscope (SEM) image of an Au nanoribbon. *I* denotes the applied current, *V* the measured voltage, and arrows indicate the current direction. Scale bar, 4 μm. **b** Schematic illustrating the Rayleigh-Plateau instability of thin ribbons. **c** Morphologies of textured, regular, and Au-Ti films with linewidths of 100 nm and 500 nm after annealing at 700 °C, 400 °C, and 600 °C, respectively. Scale bar: 1 μm. **d** Current-voltage curves of 100 nm wide ribbons after annealing at different temperatures. **e**–**g** Resistivity of textured (**e**), regular (**f**), and Au-Ti (**g**) ribbons of different widths after annealing at various temperatures. The symbol "×" indicates damaged ribbons due to Rayleigh-Plateau instability. Insets illustrate the corresponding film structures, and horizontal dashed lines indicate the bulk Au resistivity. The symbol × indicates damaged ribbons due to Rayleigh–Plateau instability. **h** Schematic diagram of a radio frequency diode device using textured Au films for electrodes and connection lines, with ground (G) and signal (S) pads arranged in a ground-signal-ground configuration. Ti metal serves as the cathode contact, palladium as the anode contact, and semiconductor single-walled carbon nanotubes as the channel layer. The lower-right corner shows the SEM morphology of the device. Scale bar, 20 μm. **i, j** Current-voltage curves of carbon nanotube diodes based on textured (**i**) and regular (**j**) films at different temperatures. Insets show SEM images of the corresponding devices after annealing at 700 °C. Scale bars: 20 μm.

EBSD analysis, atomic-resolution images show that the textured film maintains an out-of-plane (111) orientation even at triple junctions (Fig. 3f), whereas the regular film contains misoriented grains meeting at bi-junctions (Fig. 3e). Such misoriented junctions generate higher-energy boundaries that provide favorable sites for void nucleation and the onset of rupture[31].

The surface of textured films becomes markedly more planar during annealing (Fig. 3g). The strong out-of-plane (111) texture is preserved (Supplementary Fig. 13 and 14), and the grains become increasingly uniform, with few discernible defects (Fig. 3g). High-resolution imaging reveals atomically ordered structures near the graphene interface (Fig. 3h) and sharply defined vertical incoherent twin boundaries (Fig. 3i). Notably, the epitaxial interaction between the ordered Au lattice and graphene can produce moiré fringes (Supplementary Fig. 14d), underscoring the high degree of structural refinement in the annealed textured films. AFM measurements in the film interior show height fluctuations of only ~1 nm (Supplementary Fig. 15), consistent with the cross-sectional observations. The larger variations observed in the stepped-film measurements (Supplementary Fig. 6) are localized mainly near the retracting edge, where capillary-driven mass redistribution produces a less uniform surface. Unlike textured films, the high grain-boundary energy inherent in regular films leads to rupture during annealing, forming discrete nanoparticles in which numerous defects, such as dislocations and coherent twins, arise to

dissipate the strain generated during morphological evolution (Supplementary Fig. 16).

Interestingly, many tiny voids (diameter <5 nm) are already present in the as-deposited textured films (Fig. 3d), likely resulting from the incomplete coalescence during late-stage Volmer–Weber growth[26,45]. High-resolution STEM images show that the Au surrounding these voids consistently retains an out-of-plane (111) orientation (Supplementary Fig. 17). After annealing at 600 °C for 2 h, no discernible residual voids are observed by plan-view STEM in either graphene-supported textured films (Supplementary Fig. 14a) or textured films transferred onto SiO_2/Si after graphene removal (Supplementary Fig. 18), indicating that graphene-interface-induced stress is not the dominant driving force and that void healing is primarily capillarity-driven. This evolution reduces the highly curved Au-free surfaces and is mediated by surface and grain-boundary diffusion[32]. The retained Au (111) orientation at the retracted edge, together with the interrupted-annealing measurements, suggests that preservation of the textured microstructure contributes to the kinetic suppression of further edge recession (Supplementary Fig. 14e, f). Note that electron energy-loss spectroscopy shows that such metal-graphene interaction is predominantly physical in nature (Supplementary Fig. 19), providing a route for producing thermally robust, ultrathin, ultraflat, textured, and transferable metal films[46,47].

The degree of stabilization follows the ability of the growth surface to induce the textured microstructure. Multilayer graphene induces stronger Au (111) texturing, more uniform grains, and a more developed low-energy grain-boundary network than monolayer graphene (Supplementary Figs. 11 and 20), consistent with its greater post-annealing stability (Supplementary Fig. 21), whereas graphene defects weaken the templating effect and reduce both texture strength and thermal stability (Supplementary Fig. 22). In addition, this high-temperature stabilization cannot be directly generalized to other 2D materials or metals. Neither hexagonal hBN (Supplementary Fig. 5c, d) nor tetragonal Bi_2O_2Se (Supplementary Fig. 5e, f) produces a uniformly oriented Au microstructure in the examined regions, and Au films on hBN rupture upon annealing (Supplementary Fig. 5). Nanoscale voids are also present in the hBN-supported films (Supplementary Fig. 5c); however, unlike those in graphene- and MoS_2-templated films, the Au surrounding these voids does not exhibit a uniform out-of-plane orientation. This contrast suggests that local crystallographic coherence around nanoscale voids may facilitate capillarity-driven healing, whereas orientation disorder may impede healing and promote rupture. When the metal is varied instead, graphene suppresses thermally driven coarsening and roughening of Au, Cu and Pt relative to SiO_2/Si (Supplementary Fig. 23), although whether similar stabilization occurs in other metals remains to be established. These results indicate that the templating effect is selective and depends on the specific metal–template combination.

It should be noted that previous 2D-material strategies have stabilized deposited metals through mechanical confinement by capping layers[48] or enhanced adhesion via strong interfacial bonding, such as Au–Mo bonds formed after removing the top S layer of MoS_2[15]. Our result suggests a different mechanism: an underlying vdW template instead stabilizes Au by directing the formation of a strongly textured microstructure with a low-energy grain-boundary network, while leaving the metal surface unconstrained. This approach requires neither a permanent capping layer nor strong chemical bonding, and the resulting films retain their thermal stability after removal from the template, enabling ultrathin, transferable metal films that undergo levelling and nanoscale void healing rather than rupture.

Thermally robust devices in extreme environments

The extraordinary thermal stability of textured films not only creates a wide safety margin for the high-temperature annealing required in semiconductor device fabrication but also enables devices that operate reliably in extreme environments. We first demonstrate this using metal gates, which can be as narrow as ~100 nm and are therefore especially vulnerable to thermally driven instability[22]. We fabricated regular, textured, and Ti-adhered (3-nm) Au ribbons (40-nm thick) with widths ranging from 100 nm to 6 μm using electron-beam lithography. Electrical transport was characterized via four-terminal measurements (Fig. 4a).

Conventional metal ribbons fail by Rayleigh–Plateau instability upon annealing (Fig. 4b)[49]. In striking contrast, 100-nm-wide textured ribbons remain continuous (Supplementary Fig. 24), with an average width of 89.8 ± 4.5 nm and an average height of 41.4 ± 1.4 nm (Supplementary Fig. 25), after annealing at 700 °C, whereas regular ribbons rupture near 400 °C, even in the presence of a Ti adhesion layer (Fig. 4c). Electrical transport measurements across all widths confirm this structural robustness (Fig. 4d and Supplementary Fig. 26). For the same geometry, textured ribbons exhibit the lowest initial resistivity and, notably, a further drop in resistivity after annealing, approaching the bulk Au limit. Because the measurements were performed after cooling to room temperature, the reduction in resistivity persists after annealing and is therefore not a transient temperature-dependent effect. It instead reflects persistent microstructural reconstruction, including grain growth and a reduced density of grain boundaries (Supplementary Fig. 11c, d).

We further demonstrate device-level robustness using large-area CVD graphene as a texturing template for Au electrodes and interconnects. Conventional MoS_2 transistors typically lose functionality after annealing at ~400 °C due to contact failure (even when Bi or Sb adhesion layers are employed)[14]. By contrast, transistors incorporating textured Au remain operational up to 600 °C (Supplementary Fig. 27).

We also integrate textured electrodes with thermally resilient single-walled carbon nanotubes (SWNT) to form radio-frequency (RF) diodes, which can experience substantial localized Joule heating during sub-6-GHz 5 G operation (Fig. 4h). Separate transfer-length-method (TLM) measurements on textured Au–SWNT test structures show that the width-normalized Au-SWNT contact resistance decreases from 104.2 to 62.7 kΩ μm after annealing at 700 °C for 2 h (Supplementary Fig. 28). The RF diode likewise exhibits an increased on-state current after annealing (Fig. 4i), consistent with the reduced resistance of the textured Au interconnects, although annealing-induced changes in the local Ti/Pd–SWNT contacts may also contribute. These devices remain functional after five successive 2-h annealing cycles at 700 °C and after continuous annealing at 700 °C for 10 h, although prolonged exposure produces some local circuit damage (Supplementary Fig. 29). They cease to function after annealing at 800 °C for 2 h (Supplementary Fig. 29), indicating that the thermal limit of the complete device stack is lower than that of blanket textured Au films, which remain stable at 850 °C. By contrast, devices with regular electrodes degrade markedly at 400 °C owing to Rayleigh–Plateau instability (Fig. 4j). This robustness is important for device integration, where ultrathin metal contacts must retain both structural continuity and electrical functionality during thermal processing[50].

Discussion

In summary, we showed that van der Waals epitaxy can convert thermodynamically unstable ultrathin metals into thermally robust, self-healing films. Unlike conventional polycrystalline films that undergo stochastic grooving and rupture, vdW-templated films evolve toward flatter, more ordered morphologies that maintain structural continuity at temperatures close to the bulk melting point. This exceptional resilience originates from a restructured grain-boundary network dominated by low-energy, mechanically robust interfaces that suppress interior rupture and arrest further edge retraction, even in the regime of rapid surface diffusion. We therefore expect that the vdW-textured metallic films provide a scalable architecture for resilient

gates, interconnects, and contacts for the high-quality integration of 2D materials into next-generation electronics.

More broadly, the evolution of a grain-boundary groove can be understood in terms of the local capillary-force balance between the adjoining free surfaces and the grain boundary, as illustrated in Fig. 1a, b. Reducing the grain-boundary energy shifts this balance away from groove deepening and rupture and toward surface levelling and nanoscale void closure. This picture highlights how template-directed control of crystallographic texture and grain-boundary character can fundamentally redirect the thermal evolution of ultrathin metal films. A fully predictive framework incorporating anisotropic surface and grain-boundary energies and coupled surface- and grain-boundary-diffusion kinetics will be an important direction for the deterministic design of transferable, thermally robust metal films.

Methods

Preparation and transfer of textured Au film and regular Au film

Silicon wafers (1× cm^2) with a 285-nm-thick SiO_2 layer were sequentially cleaned by ultrasonication in acetone, isopropyl alcohol, and deionized water (15 min each), and then dried under N_2. The wafers were subsequently treated in an oxygen plasma cleaner (CPC-A, CIF) at 100 W for 4 min. Graphene, MoS_2, hBN, and Bi_2O_2Se crystals (Shanghai Onway Technology Co., Ltd.) were mechanically exfoliated onto the SiO_2/Si substrates to obtain mono-, bi-, tri-, or multilayer flakes using Scotch tape cleavage. A UV-ozone cleaner (Ossila) was used for the fabrication of defective graphene. Au, Cu, and Pt films were deposited onto graphene and bare SiO_2/Si substrates by electron-beam evaporation (E360) at a rate of 1 Å/s, as monitored by a quartz crystal microbalance. Before deposition, the chamber was evacuated to a base pressure below 2×10^{-8} Torr. Au deposited on graphene formed a textured film, whereas Au deposited directly on amorphous SiO_2 formed a regular film. Textured and regular Au films were transferred using a PMMA-assisted method. Briefly, PMMA was spin-coated at 2000 rpm and baked at 120 °C for 10 min to remove residual solvent. The sample was then immersed in 1 M NaOH to etch the underlying layer and release the PMMA/Au film from the growth substrate. The floating PMMA/Au film was rinsed three times in deionized water (15min each) and then scooped onto the target substrate (e.g., a TEM grid). Finally, the PMMA support was removed by rinsing in acetone. For the graphene-removal experiments on textured films, the graphene exposed on the underside of the released PMMA/Au/graphene stack was removed by oxygen-plasma etching at 50 W for 30 s. Top-view TEM specimens were prepared by transferring textured or regular Au films from SiO_2/Si onto copper grids using this PMMA-assisted method. Side-view (cross-sectional) TEM specimens were prepared by focused ion beam milling (FIB; Helios Nanolab 600i).

Annealing

Samples intended for annealing were placed on an alumina plate and loaded into a vacuum tube furnace. The tube furnace was evacuated with a mechanical pump to a base pressure below 5 Pa and held for 30 min, followed by Ar purging (50 sccm) for 20 min to minimize residual air. The furnace was then ramped at 10 °C min^{-1} to the target temperature. After annealing, the system was allowed to cool naturally to room temperature before the furnace was opened. Unless otherwise noted, the annealing times reported in this work refer to the isothermal hold at the target temperature and exclude the heating and cooling ramps.

Characterizations

The layers of graphene exfoliated on SiO_2/Si substrate were confirmed using Raman spectroscopy, which was performed with a 514 nm wavelength (Renishaw system, ~1 μm spot size, <0.5 mW power, 1 cm^{-1} spectral resolution). The topography of textured Au film and regular Au film before and after annealing was characterized using an ultra-high-resolution cold field emission scanning electron microscope (Hitachi-SU8220) and AFM (Multimode 8HR, Bruker, RTSEP-300 silicon tip). High-resolution HAADF-STEM characterization was performed using a 300 kV double spherical aberration electron microscope (Cs-STEM; FEI Titan G2 Themis). EBSD detection was achieved using a TESCAN MIRA3 LMH scanning electron microscope equipped with an Oxford-Symetrix EBSD detector, with a scanning step of 0.01 μm.

Transistor construction and thermomechanical stability measurement

Large-area single-crystal graphene was grown on copper foils by atmospheric-pressure chemical vapor deposition (APCVD) using H_2 (400 sccm) and CH_4 (2 sccm). The graphene was transferred onto SiO_2/Si substrates via a PMMA-assisted method. Device fabrication followed standard micro- and nanofabrication procedures. Briefly, PMMA 950 K electron-beam resist was spin-coated, patterned by electron-beam lithography, and developed; subsequent electron-beam evaporation of Au and lift-off yielded templated Au wirelines and device electrodes. Graphene outside the Au features was removed by oxygen plasma etching (50 W, 30 s). For the semiconducting SWNT network radio-frequency diode, Ti and Pd served as the cathode and anode contacts, respectively, and the channel consisted of a semiconducting SWNT network deposited from solution. SWNTs outside the channel region were subsequently removed by electron-beam lithography followed by reactive ion etching. For the MoS_2 field-effect transistor, MoS_2 flakes were first exfoliated onto a PDMS elastomer stamp and then transferred onto the prepatterned textured Au electrodes; P^{++} Si served as the gate electrode with 285-nm SiO_2 as the gate dielectric. Electrical measurements were carried out under vacuum on a probe station (Lakeshore TTP4) using a semiconductor parameter analyzer (Keysight B1500A).

Molecular dynamics simulations

All molecular dynamics simulations were performed with LAMMPS using the velocity-Verlet integrator with a 1 fs timestep to model physical vapor deposition of Au onto SiO_2 and graphene/SiO_2 substrates. Interactions within graphene were described by the AIREBO potential[51], Au–Au interactions by an EAM potential[52], and Si–O interactions by the Tersoff potential[53]. Nonbonded van der Waals interactions between unlike atom pairs were represented by a 12–6 Lennard–Jones potential with a 1.2 nm cutoff; corresponding parameters are provided in Supplementary Table 1. Trajectory visualization and quantitative structural analyses (local structure identification, orientation mapping, and grain segmentation) were carried out in OVITO. Amorphous SiO_2 (a-SiO_2) was generated via a melt–quench protocol starting from a cubic β-cristobalite SiO_2 supercell ($32.2 \times 32.2 \times 5$ nm^3) under periodic boundary conditions in all directions: the system was melted for 1 ns at 5000 K and 0 pressure in the NPT ensemble, quenched to 400 K at a linear cooling rate of 10^{12} K s^{-1}, and then relaxed for 100 ps in NPT at 400 K to remove residual stress, yielding structural descriptors consistent with experiments and ab initio data. The resulting bulk a-SiO_2 was then cut along the xy plane to form a 3 nm-thick slab and placed in a simulation box of $32.5 \times 31.2 \times 15$ nm^3 with periodic boundaries in x and y and a nonperiodic boundary in z; a sufficiently thick vacuum region along z accommodated incoming Au atoms. For graphene/SiO_2, a monolayer graphene sheet was stacked on the a-SiO_2 surface with an initial separation of 0.34 nm, while maintaining x–y periodicity for both layers. Each substrate was partitioned along z into a 0.5 nm fixed bottom layer to suppress rigid-body motion, a 1.0 nm thermal-control layer thermostatted (Berendsen) to maintain the substrate temperature, and an upper Newtonian region evolved without direct thermostatting and made sufficiently thick to prevent deposited Au from reaching the thermal layer. Deposition was performed with the Newtonian atoms propagated in

NVE while the thermal-control layer was maintained at 400 K (effectively NVT); Au atoms were inserted above the surface at 500 atoms ps^{-1}, with random x–y positions, a fixed insertion height of $z = 14.5$ nm, and an initial downward velocity corresponding to the deposition temperature. A repulsive wall at the top of the box prevented atom loss. After deposition, the resulting Au films were relaxed for 1 ns in NVT at 400 K, followed by dewetting simulations in which the fixed layer remained immobilized, and all other atoms were heated from 400 K to the target temperature at 5 K ps^{-1} and then equilibrated under NVT at the target temperature for 2 ns.

Data availability

The data supporting the findings of this study are provided in the main text and Supplementary Information files. Source data underlying the plots in the main text are available on GitHub and have been archived on Zenodo at https://doi.org/10.5281/zenodo.22089959. Additional data are available from the corresponding author upon request.

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Acknowledgements

Z.D. and W.W. acknowledge the support of the Peking Nanofab. W.W. thank H. Hu, Z. Song, and P. Chen of the National Center for Nanoscience and Technology for their helpful assistance with the experiments.

Author contributions

Z.D. and W.W. conceived the idea and designed the experiments. W.W. carried out the preparation and characterization of thin films. J.W. and K.Z. conducted aberration-corrected transmission electron microscopy characterization. G.W. and X.W. performed molecular dynamics simulations. W.W., C.J. and S.Z. performed high-temperature annealing. W.W., C.Y., T.S., J.C. and Z.D. analyzed the data. W.W. and Z.Y. fabricated the radio-frequency diode device. Z.D. and W.W. wrote the manuscript. All the authors reviewed and approved the manuscript.

Funding

This work is supported by the National Natural Science Foundation of China (grant nos. 12372103 and 12432003) and the Natural Science Foundation of Beijing (grant no. 2264104). W.W. acknowledges the support of the Postdoctoral Science Foundation (grant no. GZC20251279). T.S. acknowledges support from the European Research Council (ERC) through the EMetBrown ERC Consolidator Grant (grant no. 101039103).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-77794-5.

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Peer review information *Nature Communications* thanks Lingan Kong, Pan Wang and Peijian Wang for their contribution to the peer review of this work. A peer review file is available.

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